Limits – 3

Asymptotics of sequences

Homework

1 сентября 2026 года

Theoretical minimum. The lemma on a neutral fixed point $x_{n+1} = x_n - ax_n^p + o(x_n^p)$; the Stolz–Cesàro theorem; telescoping in powers of x_n; comparison of a sum with an integral; the asymptotics $H_n = \ln n + \gamma + O(1/n)$; the step from "the limit is zero" to the leading term together with an estimate of the remainder.

---

1. Let $x_0 \in [-1, 1]$ and $x_{n+1} = x_n - \arcsin(\sin^3 x_n)$ for $n \geqslant 0$. Compute $\lim_{n\to\infty} \sqrt{n}\, x_n$.

2. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and suppose the sequence $a_n = \int_0^1 f(n + x)\, dx$ converges. Prove that the sequence $b_n = \int_0^1 f(nx)\, dx$ converges as well.

3. (ФКН, осень 2023) A sequence of points of the plane is given by $(x_1, y_1) = (0, 0)$ and

$$(x_{n+1}, y_{n+1}) = \Big(\Big(1 - \frac{2}{n}\Big)x_n - \frac{1}{n}y_n + \frac{4}{n},\ \Big(1 - \frac{1}{n}\Big)y_n - \frac{1}{n}x_n + \frac{3}{n}\Big).$$

   Prove that the limit $\lim_{n\to\infty}(x_n, y_n)$ exists, find it, and show that the distance from (x_n, y_n) to the limit is $O\big(n^{-(3-\sqrt{5})/2}\big)$.

4. (IMC 2025 shortlist) Compute $\lim\limits_{n\to\infty} \dfrac{n^1 + n^2 + \cdots + n^n}{1^n + 2^n + \cdots + n^n}$.

5. (SEEMOUS) Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous strictly decreasing function with $f([0, 1]) \subseteq [0, 1]$.

   a) Prove that for every $n \geqslant 1$ the equation $f(x) = x^n$ has a unique solution $a_n \in (0, 1)$, and that $\lim_{n\to\infty} a_n = 1$.

b) Suppose in addition that f has a continuous derivative, $f(1)=0$ and $f'(1)<0$, and put $F(x)=\int_x^1 f(t)\,dt$. For which $\alpha\in\mathbb{R}$ does the series $\sum_{n\geqslant 1}\big(F(a_n)\big)^{\alpha}$ converge?

6. (IMC 2024) Define $x_1=2$, $x_2=4$ and $x_{n+2}=3x_{n+1}-2x_n+\frac{2^n}{x_n}$ for $n\geqslant 1$. Prove that $\lim_{n\to\infty}\frac{x_n}{2^n}$ exists, and estimate it.

7. (OMOUS 2024) For an integer $n\geqslant 2$ let $I(n)=\displaystyle\int_0^{\pi/2}\frac{\{nx\}}{\sin x}\,dx$, where $\{s\}$ is the fractional part of s. Prove that the integral converges, that $C_1\ln n\leqslant I(n)\leqslant C_2\ln n$ for some constants $C_1,C_2>0$, and find $\lim_{n\to\infty}\frac{I(n)}{\ln n}$.

8. (Vienna MC 2025) Let (a_n) satisfy $a_0>0$ and $a_{n+1}=\frac{a_n}{a_n^2+a_n+1}$ for all n. Show that $a_n\to 0$ and determine $\lim_{n\to\infty} n a_n$.

9. (SEEMOUS) For $n\geqslant 1$ put

$$x_n=\int_0^1 \ln\big(1+x+x^2+\cdots+x^n\big)\cdot\ln\frac{1}{1-x}\,dx.$$

a) Show that x_n is finite for every n and that $\lim_{n\to\infty}x_n=2$.

b) Compute $\lim_{n\to\infty}\frac{n}{\ln n}\,(2-x_n)$.

10. (VJIMC 2002) Prove that $\displaystyle\lim_{n\to\infty} n^2\Big(\int_0^1\sqrt[n]{1+x^n}\,dx-1\Big)=\frac{\pi^2}{12}$.